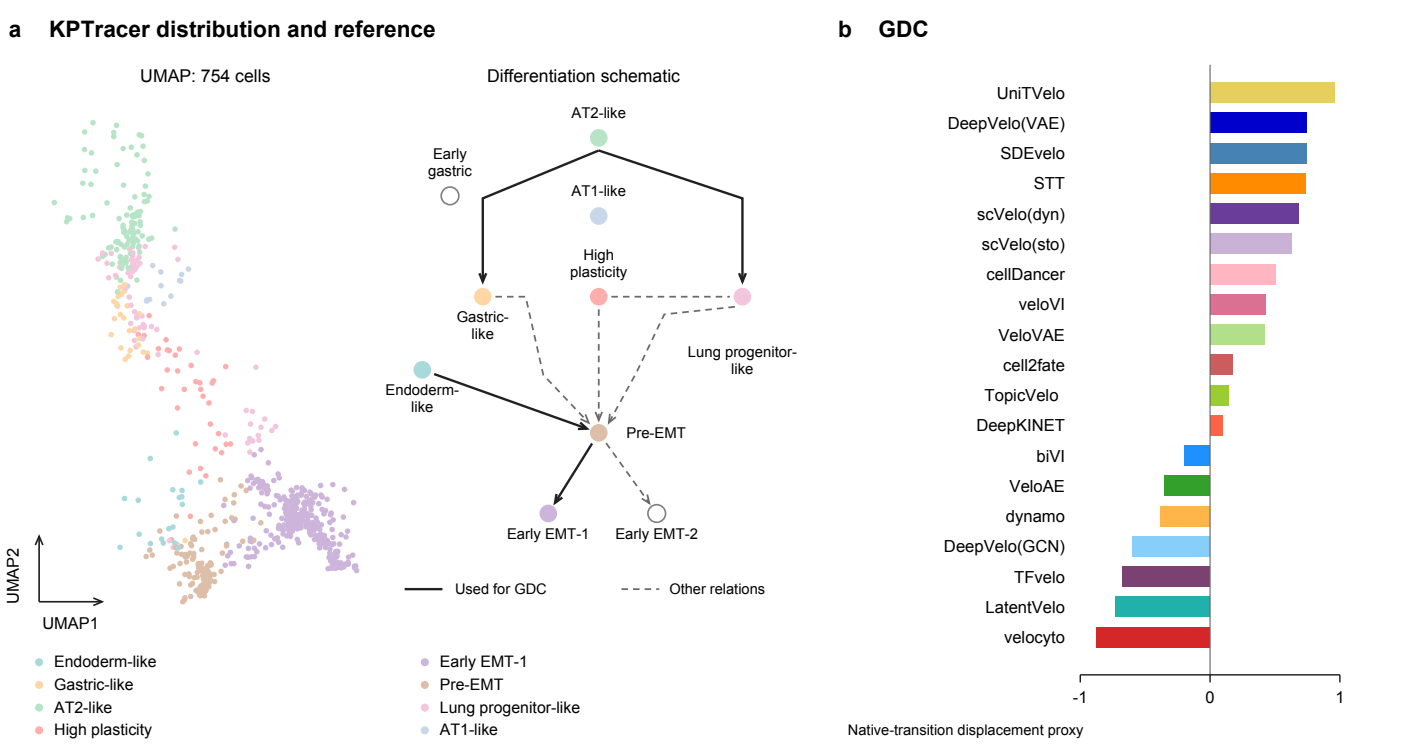
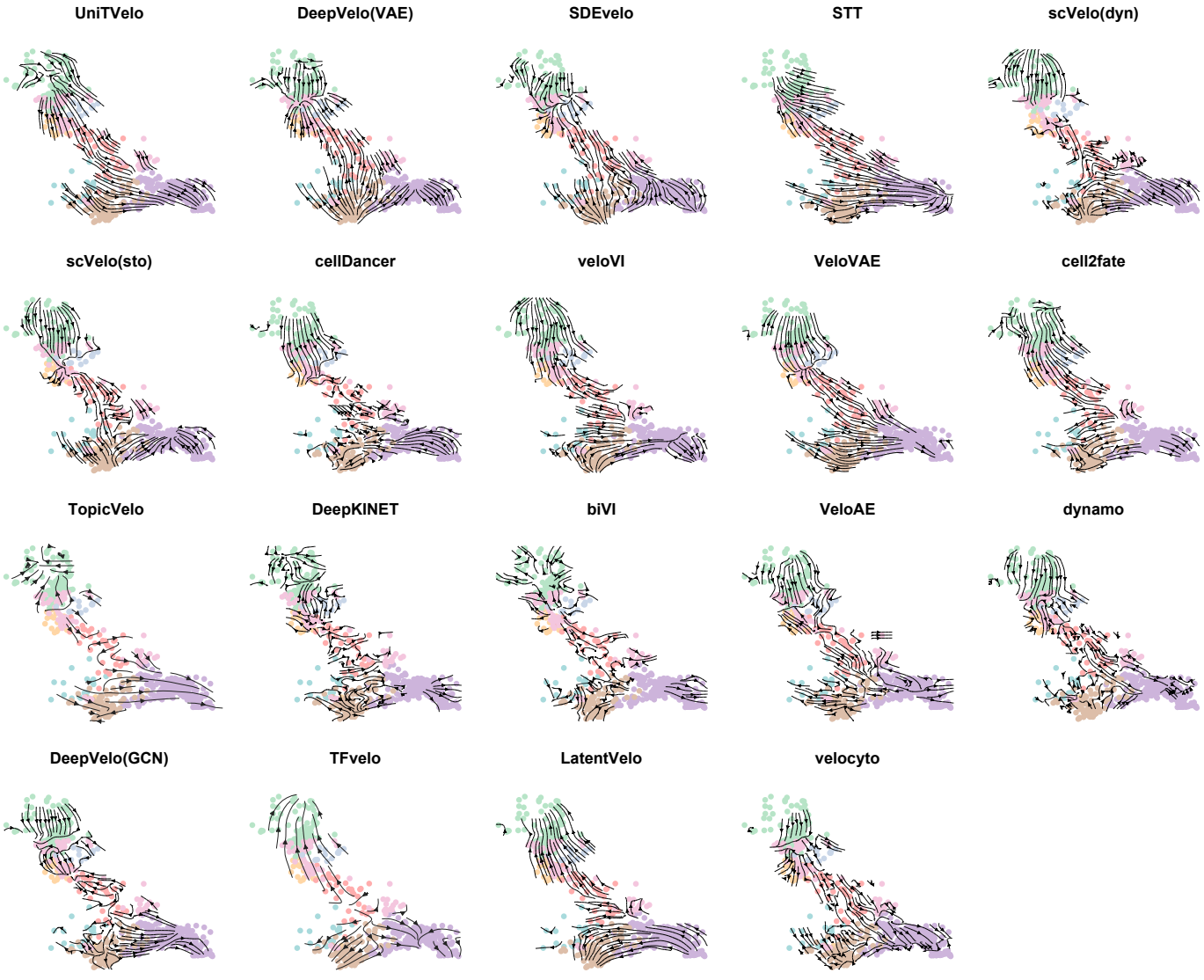


c Method-specific velocity fields



Extended Data Fig. 11 | KPTracer: reference, scores and method fields

a, All 754 KPTracer cells in the fixed UMAP, with UMAP1/UMAP2 axes and source cell-type colors. The schematic uses the plain node-and-arrow style of Extended Data Fig. 9, without background regions, and is adapted from Yang et al. (Cell 2022, doi:10.1016/j.cell.2022.04.015), Fig. 5I. Solid arrows mark the four relations used for GDC; dashed connections show other literature relations and are excluded from GDC. The dashed horizontal connection denotes an undirected coupling. Line style encodes GDC inclusion, not the original evidence categories. Hollow nodes are literature states absent from this subset. Unconnected mesenchymal states are omitted for clarity. The source Gastric/Endoderm progression is expanded into the evaluated Endoderm relation and a contextual Gastric branch; Early EMT-1/2 is split to identify the evaluated EMT-1 state. Node positions are schematic, not UMAP coordinates or measured trajectories. Historical Lung Mixed labels correspond to Lung progenitor-like. b, GDC ranked in descending order, using the 706 cells in the six reference-participating states as denominator. The 17 historical corrected eval1 scores and the completed TFvelo/TopicVelo scores are unchanged. TopicVelo denotes a native-transition displacement proxy: within-state coherence uses T @ Ms - Ms; boundary directions use native T projected into the fixed UMAP, without historical base-scVelo velocity. c, Nineteen method fields in the same order as b, read left to right and then top to bottom. Saved streamline paths are aligned by per-panel affine transforms to identical cell-coordinate bounds, with uniformly enlarged cell markers. Panel a also uses enlarged cell markers. TFvelo uses 1,950 finite fitted genes. TopicVelo resumes topic 7 of 8 from the saved fit at the 35th-percentile threshold (490 topic cells; 727 nonzero projected directions), retaining the native package fallback. Gene sets and preprocessing differ by method. These are descriptive results from one dataset; frozen main Fig. 6 composite inputs remain unchanged.